

Algebra 1: Polynomials

Quiz Level

Name: _____

Date: _____

Question 1: Simplify the expression $x^2 + 3x + 2$

- A) $x^2 + 5x + 6$
- B) $x^2 + 3x + 2$
- C) $x^2 + 2x + 1$
- D) $x + 2$
- E) $2x + 1$

Question 2: Factor the polynomial $x^2 - 4$

- A) $x^2 - 2x + 2x - 4$
- B) $x^2 + 4$
- C) $(x - 2)(x + 2)$
- D) $(x + 2)(x + 2)$
- E) $(x - 2)(x - 2)$

Question 3: Solve for x : $2x^2 + 5x - 3 = 0$

- A) $x = -3$ or $x = 1$
- B) $x = -1$ or $x = 3$
- C) $x = 1$ or $x = -1$
- D) $x = 1$
- E) $x = -1$

Question 4: Find the roots of the equation $x^2 - 2x - 3 = 0$ using the quadratic formula

- A) $x = -1$ or $x = 3$
- B) $x = 1$ or $x = -3$
- C) $x = 3$ or $x = -1$
- D) $x = -3$
- E) $x = 1$

Question 5: Graph the polynomial $y = x^3 - 2x^2 - 5x + 6$ and identify its x-intercepts

- A) $(-2, 0)$, $(1, 0)$, $(3, 0)$
- B) $(-3, 0)$, $(1, 0)$, $(2, 0)$
- C) $(-1, 0)$, $(2, 0)$, $(3, 0)$
- D) $(-2, 0)$, $(-1, 0)$, $(3, 0)$
- E) $(-1, 0)$, $(1, 0)$, $(2, 0)$

Question 6: If $f(x) = x^2 + 2x - 6$, find $f(-2)$

- A) 0
- B) -2
- C) -6
- D) 2
- E) -12

Question 7: Simplify the expression $(x^2 + 3x + 2) + (x^2 - 2x - 1)$

- A) $2x^2 + x + 1$
- B) $2x^2 + x + 3$
- C) $x^2 + 2x + 1$
- D) $2x^2 - x - 3$
- E) $x^2 + x - 3$

Question 8: Prove that $(a + b)(a - b) = a^2 - b^2$ using algebraic manipulation

Question 9: Solve the equation $x^3 - 2x^2 - 5x + 6 = 0$ by factoring or using synthetic division

Question 10: For the polynomial $y = 2x^3 - 5x^2 + x + 1$, find its first and second derivatives, y' and y''

Chef's Tips (Hints):

- To simplify expressions, start by combining like terms
- When solving quadratic equations, try factoring first, then use the quadratic formula if necessary
- To graph polynomials, identify x-intercepts by solving $y = 0$ and consider end behavior